

TASK 1

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <pthread.h>
```

```
#include <unistd.h>
```

```
#include <time.h>
```

```
#define BUFFER_SIZE 1000
```

```
#define TOTAL_DATA 100000
```

```
int buffer[BUFFER_SIZE];
```

```
int in = 0;
```

```
int out = 0;
```

```
int count = 0;
```

```
pthread_mutex_t mutex = PTHREAD_MUTEX_INITIALIZER;
```

```
pthread_cond_t not_full = PTHREAD_COND_INITIALIZER;
```

```
pthread_cond_t not_empty = PTHREAD_COND_INITIALIZER;
```

```
void *producer(void *arg)
```

```
{
```

```
    for (int i = 0; i < TOTAL_DATA; i++)
```

```
    {
```

```
        pthread_mutex_lock(&mutex);
```

```
        while (count == BUFFER_SIZE)
```

```
            pthread_cond_wait(&not_full, &mutex);
```

```

    buffer[in] = i;
    in = (in + 1) % BUFFER_SIZE;
    count++;

    pthread_cond_signal(&not_empty);
    pthread_mutex_unlock(&mutex);
}

printf("Producer finished.\n");

return NULL;
}

void *consumer(void *arg)
{
    for (int i = 0; i < TOTAL_DATA; i++)
    {
        pthread_mutex_lock(&mutex);

        while (count == 0)
            pthread_cond_wait(&not_empty, &mutex);

        int data = buffer[out];
        (void)data;

        out = (out + 1) % BUFFER_SIZE;
        count--;
    }
}

```

```

        pthread_cond_signal(&not_full);
        pthread_mutex_unlock(&mutex);
    }

    printf("Consumer finished.\n");

    return NULL;
}

int main()
{
    pthread_t producer_thread;
    pthread_t consumer_thread;

    struct timespec start, end;

    clock_gettime(CLOCK_MONOTONIC, &start);

    pthread_create(&producer_thread, NULL, producer, NULL);
    pthread_create(&consumer_thread, NULL, consumer, NULL);

    pthread_join(producer_thread, NULL);
    pthread_join(consumer_thread, NULL);

    clock_gettime(CLOCK_MONOTONIC, &end);

    double communication_time =
        (end.tv_sec - start.tv_sec) +
        (end.tv_nsec - start.tv_nsec) / 1e9;

```

```

double throughput = TOTAL_DATA / communication_time;

printf("Data produced : %d\n", TOTAL_DATA);
printf("Communication time : %.6f seconds\n",
       communication_time);
printf("Communication throughput : %.2f values/second\n",
       throughput);

pthread_mutex_destroy(&mutex);
pthread_cond_destroy(&not_full);
pthread_cond_destroy(&not_empty);

return 0;
}

```

```

nikith@NIKITH:~$ mkdir -p ~/ossp
cd ~/ossp
nikith@NIKITH:~/ossp$ pwd
/home/nikith/ossp
nikith@NIKITH:~/ossp$ nano producer_consumer.c
nikith@NIKITH:~/ossp$ gcc producer_consumer.c -o producer_consumer -pthread
nikith@NIKITH:~/ossp$ ./producer_consumer
Consumer finished.
Producer finished.
Data produced : 100000
Communication time : 0.027604 seconds
Communication throughput : 3622623.09 values/second
nikith@NIKITH:~/ossp$ |

```

TASK 2

```

#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/wait.h>

```

```
int main()
{
    int pipefd[2];

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        return 1;
    }

    pid_t pid1 = fork();

    if (pid1 < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid1 == 0)
    {
        // Child 1: execute ls -l

        close(pipefd[0]);

        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[1]);

        execlp("ls", "ls", "-l", NULL);
```

```
    perror("execlp ls");  
    exit(1);  
}
```

```
pid_t pid2 = fork();
```

```
if (pid2 < 0)
```

```
{  
    perror("fork");  
    return 1;  
}
```

```
if (pid2 == 0)
```

```
{  
    // Child 2: execute grep ".c"
```

```
    close(pipefd[1]);
```

```
    dup2(pipefd[0], STDIN_FILENO);
```

```
    close(pipefd[0]);
```

```
    execlp("grep", "grep", ".c", NULL);
```

```
    perror("execlp grep");  
    exit(1);  
}
```

```
// Parent process
```

```
close(pipefd[0]);
```

```
close(pipefd[1]);
```

```
waitpid(pid1, NULL, 0);
```

```
waitpid(pid2, NULL, 0);
```

```
return 0;
```

```
}
```

```
nikith@NIKITH:~$ cd ~/ossp
nikith@NIKITH:~/ossp$ nano ls_grep.c
nikith@NIKITH:~/ossp$ gcc ls_grep.c -o ls_grep
nikith@NIKITH:~/ossp$ ./ls_grep
-rw-r--r-- 1 nikith nikith 1044 Sep 21 05:07 ls_grep.c
-rwxr-xr-x 1 nikith nikith 16816 Sep 21 05:05 producer_consumer
-rw-r--r-- 1 nikith nikith 2160 Sep 21 05:04 producer_consumer.c
nikith@NIKITH:~/ossp$ |
```